

Detailed Summary of Churn Analysis (Percentage Based)

Based on the exploratory data analysis and visualizations, several important customer churn patterns were identified across demographics, contract type, service usage, and payment behaviour.

Demographic Insight (Senior Citizens)

Senior citizens show a **higher churn percentage** compared to non-senior customers. The churn rate among senior citizens is noticeably higher (around **40–45%**) compared to non-senior customers (around **25–30%**). This indicates that senior citizens may require better service support or targeted retention strategies.

Tenure-Based Insight

Customers with very low tenure (especially **0–2 months**) show the **highest churn percentage (~50–60%)**, indicating poor onboarding or unmet expectations. Customers with tenure greater than **24 months show very low churn (~10–15%)**, suggesting strong loyalty once customers stay longer.

Contract Type Insight

Contract type is one of the strongest churn indicators:

- **Month-to-Month Contracts → Highest churn (~40–45%)**
- **One-Year Contracts → Moderate churn (~15–20%)**
- **Two-Year Contracts → Lowest churn (~5–10%)**

This shows that long-term contracts significantly reduce churn risk.

Internet Service Insight

Churn varies significantly across internet types:

- **Fiber Optic → Highest churn (~40%+)**
- **DSL → Moderate churn (~20–25%)**
- **No Internet → Lowest churn (~5–10%)**

Fiber customers may be price-sensitive or have higher service expectations.

Service Add-On Insight (Very Important Business Finding)

Customers **without additional services** show higher churn:

Service	Churn Without Service	Churn With Service
Online Security	~40%	~15–20%
Online Backup	~35%	~20%
Device Protection	~35–40%	~20%

This suggests bundling add-on services can improve retention.

Payment Method Insight

Payment behaviour also impacts churn:

- **Electronic Check → Highest churn (~45%)**
- **Mailed Check → Moderate churn (~20–25%)**
- **Bank Transfer / Credit Card → Lowest churn (~15–20%)**

Electronic check users may have lower engagement or higher billing friction.

Monthly Charges Insight

Customers with **higher monthly charges (>70)** show higher churn percentage compared to lower-paying customers. High-cost plans may require better value communication or service quality.

Overall Business Conclusion

Customer churn is strongly driven by:

- Contract flexibility
 - Tenure duration
 - Internet service type
 - Add-on service adoption
 - Payment behaviour
 - Monthly pricing
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Strategic Business Recommendation

To reduce churn, the company should:

- Encourage long-term contracts
 - Bundle security and protection services
 - Improve onboarding for new customers
 - Review fiber service pricing and quality
 - Promote auto-payment methods
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One-Line Executive Summary

Customer churn risk is highest among **new customers, month-to-month contract users, fiber internet users, electronic check payers, and customers without add-on services**, with churn percentages ranging from **35% to 60%** in high-risk segments.